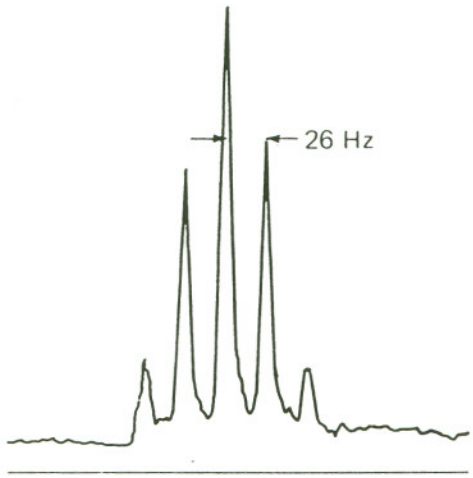
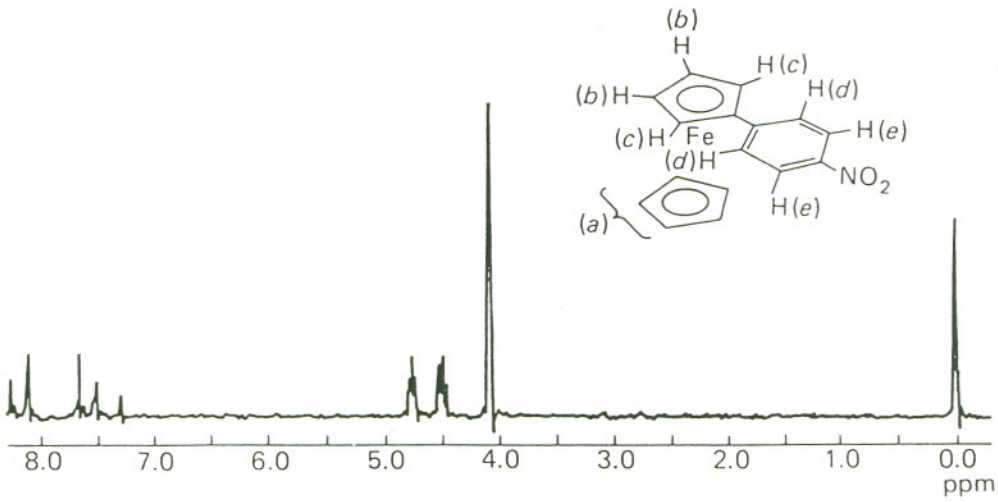
INORGANIC CUMULATIVE EXAMINATION
October 3, 1992

1. The proton resonance of the hydridic hydrogen of $HNi[OP(C_2H_5)_3]_4^+$ is given as follows. Propose a structure for this complex. (Ni, I = 0)

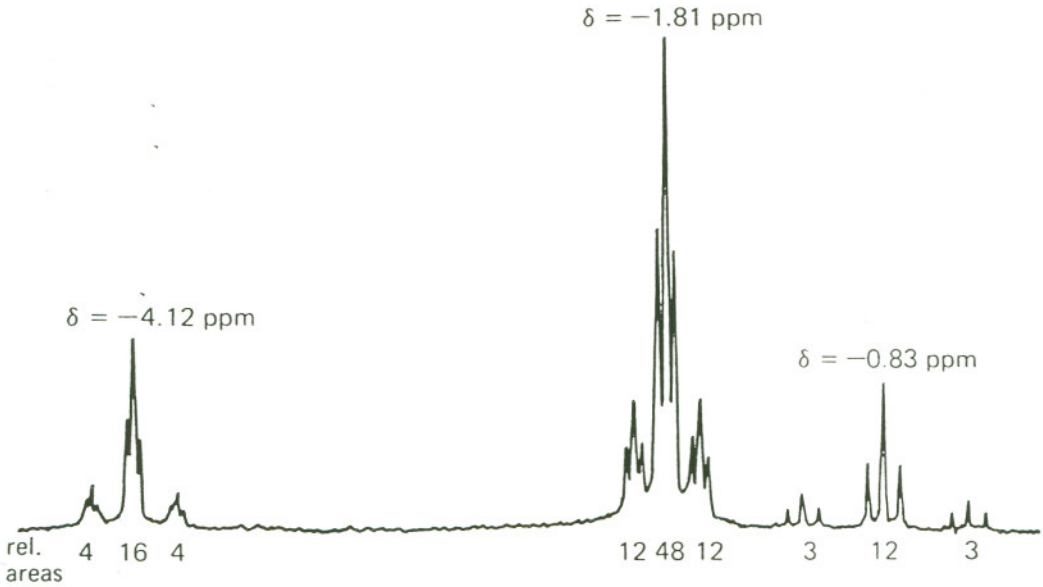


2. Given the following proton NMR spectrum and molecular structure, assign the peaks. You need not distinguish between similar protons i.e., b and c.

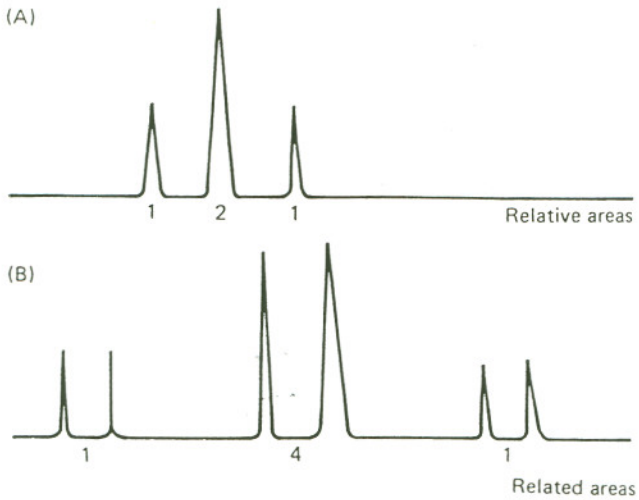


3. Interpret the spectrum on the next page for the compound $CH_3Pt(CH_2{=}CH_2)[PC_6H_5(CH_3)_2]_2$. Justify all splittings. Is this the *cis* or *trans* isomer? (*Hint*: For ^{31}P, I = 1/2, abundance = 100%. For ^{195}Pt, I = 1/2, abundance = 34%.) [See Inorg. Chem., 12, 994 (1973).] "*See Hint at Bottom of Exam*"

3. Cont....



4. Spectrum (a) results for the methyl region of one isomer of $Pd[P(CH_3)_2C_6H_5]_2I_2$. Spectrum (b) is obtained in the methyl region for the other isomer of $Pt[P(CH_3)_2C_6H_5]_2I_2$. Refer to problem 3 for background information. Note that one isomer is a Pd complex (I = 0) and the other is a Pt complex.



a) Which is *cis* and which is *trans*?

b) Explain why the areas in (b) are in the ratio of 1 to 4 to 1.

HINT: A very interesting result is obtained in the proton or carbon-13 nmr spectra of phosphorus complexes. In the proton nmr of *trans*-$PdI_2[P(CH_3)_3]_2$, we might expect to find a doublet methyl resonance with perhaps a small splitting of each peak from the second phosphorus. Although the two phosphorus atoms are *chemically equivàlent (isochronous)* they are not *magnetically equivalent*. Any given methyl group would experience two different phosphorus couplings, $^2J_{P-H}$ and $^4J_{P-H}$. When the $^2J_{P-P}$ is much larger than $^2J_{P-H}F$, as it is in the above complex, the proton nmr spectrum observed is a 1:2:1 triplet; i.e., the two phosphorus nuclei behave as though they were two equivalent nuclei splitting the proton resonance. The two phosphorus nuclei are said to be *virtually coupled*. In a similar *cis* complex where $^2J_{P-P} << {}^2J_{P-H}$, a doublet is obtained. In view of the very common, large $^2J_{P-P}$ for *trans* phosphorus ligands and the small $^2J_{P-P}$ values for *cis* phosphines, this phenomenon can be used to distinguish these two kinds of isomers.